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WEB-ENABLED AND USER-FRIENDLY PYTHON PIPELINE FOR REE MAPPING FROM HYPERSPECTRAL IMAGERY

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Overview

1. Context

2. Architecture

3. Results

4. Conclusions

Context and Positioning

- Key idea hybrid physics + ML method for Nd detection
- Focus on methodological performance and validation
- This thesis: operationalization of the method
- From algorithm $\rightarrow$ user-friendly processing platform

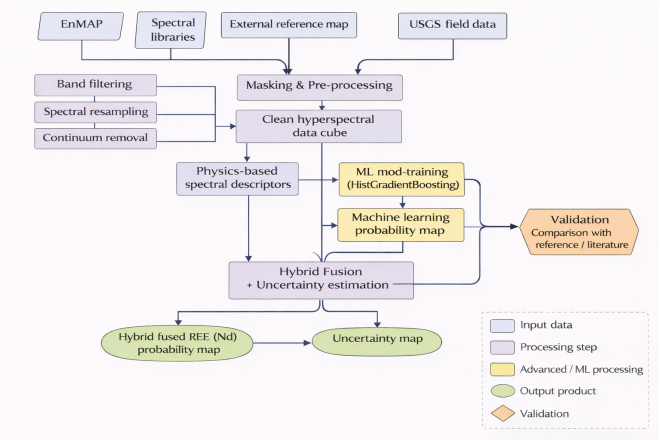
Thesis Contribution

- Modular Python processing framework following SWE paradigms
- Integrated physics-based modelling and ML inference
- Parallelized processing
- Interactive, user-friendly web interface (Streamlit)

UML-like module diagram of the HMR pipeline (modules and functions)

Platform Architecture

- End-to-end hyperspectral processing pipeline
- Data ingestion (ENVI / GeoTIFF)
- Spectral feature extraction and pseudo-labeling
- ML ensemble and hybrid fusion
- Georeferenced outputs for GIS integration



Methods:

- Continuum removal and polynomial band-shape modelling
- Physically interpretable absorption descriptors (depth, area, position)
- Physics-guided pseudo-label generation
- Calibrated ensemble learning (HistGradientBoosting)
- Hybrid fusion in logit space with uncertainty estimation

i.e. Nd diagnostic windows analyzed via continuum removal and polynomial fitting

Results: Hybrid vs Physics and ML

- Physics-based detection -> spectrally meaningful but noisy
- ML-only detection -> spatially smooth but less interpretable
- Hybrid fusion -> sharp, coherent, and physically consistent
- Strong agreement with published Nd anomalies

Conclusions and Outlook

- Tool for exploration, monitoring, and decision support
- The platform bridges theoretical research and operational EO use
- Ready for CHIME and IRIDE missions
- Extendable to other geological contexts

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Thank you for your attention

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